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Associations Between the Nurse Practice Environment and Nurse-Sensitive Patient Outcomes: Empirical Research Quantitative

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ABSTRACT

Aim: This study aimed to investigate associations between the nurse practice environment and nurse-sensitive patient outcomes in Hong Kong.

Design: This was a 4-wave repeated cross-sectional study.

Methods: The sample comprised 71 adult general medical, surgical and orthopaedic wards recruited from four public hospitals. Data on nurses' practice environment and nurse-sensitive patient outcomes were collected quarterly from each hospital over a one-year period from January 2016 to December 2018. The number of participating nurses per survey wave ranged from 604 to 865. Measures comprised the Practice Environment Scale of the Nursing Work Index and four nurse-sensitive patient outcomes identified from the National Quality Forum-endorsed nursing-sensitive care measure set. The associations between each outcome and the practice environment were examined using multiple regression analysis.

Results: Nurse practice environments were rated favourably. The 'Nursing foundations for quality of care' subscale received the highest score (59.6) while the 'Staffing and resource adequacy' subscale received the lowest score (36.7). Though no significant associations were found between the overall practice environment and nurse-sensitive patient outcomes, the subscale of 'Staffing and resource adequacy' was significantly associated with the incidence of inpatient falls and catheter-associated urinary tract infections.

Conclusions: Our findings suggest that staffing is a core nurse workplace concern which impacts the outcomes of inpatient falls and catheter-associated urinary tract infections. However, the overall nurse practice environment was not significantly associated with any patient outcomes, and further studies are needed to corroborate these results. Additional mediating factors such as nurse resilience, which may potentially affect patient outcomes, should also be explored.

No Patient or Public Contribution: Patients or the general public were not involved in the design, analysis, or interpretation of the data in this study.

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1 | Introduction

In recent decades, evidence-based nursing practice has become a fundamental approach towards improving patient health and safety (Mackey and Bassendowski 2017; Youngblut and Brooten 2001). In particular, the development of nurse-sensitive patient outcomes, which measure the effect of nursing care on patient outcomes, has facilitated extensive research and identified associations between nursing practice and patient outcomes, thus promoting advancements in healthcare policy (Burston et al. 2014; Oner et al. 2021).

The relationship between nurses' care quality and patient outcomes is moderated by aspects of the nurse practice environment, with poorer work conditions linked to negative patient experiences and increased frequency of adverse events (Mohr et al. 2013). The environment in which patients receive care is one of the four metaparadigms of nursing emphasised in several foundational and transformative theories including Nightingale's Environmental Theory (Neuman 1996) and Neuman's Systems Model (Zborowsky 2014), both of which foreground the role of nurses in identifying and mediating environmental impacts on patient health. As nurses themselves also constitute part of the environment, workplace factors such as workload and staffing policies can have significant effects on care delivery and lead to adverse patient outcomes. Notably, with the growing global focus on nursing shortages and safe and effective care, nurse to patient ratios have been determined as a core nursing factor affecting several nurse-sensitive patient outcomes such as hospital-acquired infections, fall incidence and pressure injuries (Buchan and Catton 2023; Driscoll et al. 2018).

In addition to nurse staffing, other dimensions of the nurse practice environment affecting care quality have been identified through extensive investigations with the Practice Environment Scale of the Nursing Work Index (PES-NWI) (Lake 2002; Swiger et al. 2017). By quantifying fundamental features of the work environment which influence nursing behaviours, the scale has been instrumental in identifying limitations of existing healthcare policies and supporting the professionalisation of nursing practice (Lundmark 2008). However, despite the abundance of studies investigating the effects of nurses' work conditions on patient health, the evidence remains mixed and warrants further investigation. Regarding the outcome of pressure injuries, for instance, a systematic review of 29 studies found conflicting results regarding the relationship with nurse staffing, with both negative and positive associations reported (Stalpers et al. 2015). Similarly, another recent systematic review found mixed evidence on correlations between medication errors, hospital-acquired infections and the practice environment, with variances in findings being attributed to study methods, such as diversity in operational definitions of patient outcomes, research settings and data sources (Al-Ghraiybah et al. 2021).

Moreover, in terms of regional evidence, while nurse staffing has been a recurring concern with potential implications for patient outcomes in Hong Kong (Leung et al. 2020), less attention has been paid towards evaluating the nurse practice environment as a whole. Though nurses' work experiences and turnover intention in the region have been well-documented (Choi et al. 2022; Hung and Lam 2020), to date there remain no studies conducting standardised measurements of the practice environment and its potential associations with nurse-sensitive patient outcomes. Considering nursing shortages, workplace-associated burnout and the consequent deterioration in nurses' care abilities, it is imperative to strengthen the evidence to support optimised nursing practice.

2 | Methods

2.1 | Aim

This study aimed to examine the associations between nurses' perception of the practice environment in four major hospitals and four key nurse-sensitive patient outcomes.

2.2 | Design and Sampling

This investigation follows the methods and forms part of a larger study examining the association between the nurse-to-patient ratio, nurse practice environment and nurse-sensitive patient outcomes in acute hospitals in Hong Kong (Chau et al. 2025). This was a 4-wave repeated cross-sectional survey conducted quarterly in four hospitals from January 2016 to December 2018. A convenience sampling method was employed to select four hospitals from a total of 18 public hospitals offering 24-h emergency services in Hong Kong. The adult medical, surgical and orthopaedic wards of the selected hospitals, comprising a total of 71 wards, were included. Invitations to participate in the study were sent to all full-time registered nurses (RNs) working in the wards with direct patient care duties accounting for over half of their job responsibilities. Direct patient care duties are defined as assessing and monitoring patients, administering treatments such as medications to them, communicating with them and their families, providing health education and coordinating, planning, and evaluating their care.

2.3 | Measures and Source of Data

2.3.1 | Nurse Practice Environment

The Practice Environment Scale of the Nursing Work Index (PES-NWI) is a validated scale that has been used extensively in research on nurses' working conditions and their impact on both nurse and patient outcomes (Al-Ghraiybah et al. 2021). The 31-item scale is divided into five subscales as follows: (Mackey and Bassendowski 2017) Nurse participation in hospital affairs; (Youngblut and Brooten 2001) Nursing foundations for quality of care; (Burston et al. 2014) Nurse manager ability, leadership and support of nurses; (Oner et al. 2021) Staffing and resource adequacy; and (Mohr et al. 2013) Collegial nurse-physician relations. Each item is rated on a 4-point Likert scale from 1 (Strongly disagree) to 4 (Strongly agree). The subscale and composite scores are calculated by summing up the item scores and deriving the average. The lowest possible score is 1 and the highest possible score is 4. For the purposes of the current study, scoring was rescaled to the range of 0–100 to facilitate the interpretation of association analysis results such that a one-unit increase in the

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score corresponds to a 1% increase in the score. Accordingly, a score ≥ 50 indicates a favourable working environment and a score ≤ 49 indicates an unfavourable work environment. For this study, the Chinese version of the PES-NWI, which had good overall psychometric and a Cronbach's alpha of 0.97 for the total scale, was adopted (You et al. 2012).

2.3.2 | Nurse-Sensitive Patient Outcomes

Monthly data were collected on the frequency of the following outcomes in each participating ward: (Mackey and Bassendowski 2017) pressure injuries (hospital-acquired), (Youngblut and Brooten 2001) inpatient falls, (Burston et al. 2014) physical restraints and (Oner et al. 2021) catheter-associated urinary tract infections (CAUTIs) in accordance with the operating definitions of the selected nurse-sensitive patients outcomes documented in the implementation guide for the National Quality Forum-endorsed nursing-sensitive care measure set (The Joint Commission 2009). The data management systems of participating hospitals, including the Clinical Management System (CMS), Clinical Data Analysis and Reporting System (CDARS), paper-based patient records and nursing documentation, were used to collect and compile the required data.

2.4 | Data Collection

Data collection took place over a 12-month period in each hospital. Regarding nurses' perceptions of the work environment, hard copies of the PES-NWI were distributed to the study wards at 3-month intervals. Boxes were placed in the wards to facilitate participants in returning the surveys. Regarding the collection of data on nurse-sensitive patient outcomes, extensive discussions were held with participating hospitals to achieve consensus on the data collection procedures. Primarily, relevant data were extracted on a monthly basis from the CMS, CDARS and review of patient medical records and nursing documentation. As data on CAUTIs and physical restraint use were not recorded in the CMS, these outcomes were charted daily by the research team as agreed with the relevant hospitals. A research nurse and three research assistants were recruited and trained to support data collection in the wards. Regarding CAUTIs, nurses from the Infection Control Team of each hospital, which conducts surveillance of hospital-acquired infections, were also consulted for data collection. To ensure data accuracy, the co-investigators conducted a random audit of the collected data and no errors were detected.

2.5 | Ethical Consideration

Ethical approval was obtained from the relevant university and hospital ethics committees. All participants received an explanation of the study and provided written informed consent to join the study. Participants' rights were protected according to institutional policies, local regulations, the Declaration of Helsinki, the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Guideline for Good Clinical Practice (GCP), and compliance with the Hong Kong Personal Data (Privacy) Ordinance. The surveys for nurses were conducted anonymously and approval to access patient records and nursing care records was received from participating hospitals. All collected data were stored securely by the principal investigator in encrypted files and patient identifiers were removed from the data and re-labelled in order to ensure privacy. All data analysis was done anonymously.

2.6 | Statistical Analyses

Data were summarised and presented using appropriate descriptive statistics. The normality of variables with continuous data was assessed using skewness and kurtosis statistics. Skewness and kurtosis values within −2 to 2 and −7 to 7 respectively indicate the underlying variable is not deviated from normal distribution (Curran et al. 1996). The nurse-sensitive patient outcomes of interest of the study were pressure injuries, fall incidence, physical restraint and CAUTIs. As the patients were not solely cared for by specific nurses, there was no pairing or hierarchical clustering of patients nested within nurses. The nurse-sensitive outcomes were also not repeatedly collected from the same patients over time. Average monthly data of the study outcomes collected over 12 months in each participating ward were calculated to quantify their average levels spanning over the 12-month study period.

The explanatory variable of interest of the study was the practice environment. The average composite score of the PES-NWI collected from all nurse participants in the four tri-monthly surveys in a participating ward was used to quantify the average level of practice environment in the ward spanning over the 12-month study period. The unit of analysis was at ward level. Multiple regression was used to examine the association between each of the average levels of nurse-sensitive patient outcomes and the average level of practice environment over the 12-month study period at the ward level. Potential effects attributing to differences in hospitals were adjusted as a covariate in each of the multiple regression models for each of the outcomes and the practice environment. The outcomes of the number of patients with pressure injuries, fall incidence and the number of patients with CAUTIs were all positively skewed and, therefore, square root transformed before being entered into the regression models. All statistical analyses were performed using IBM SPSS 26.0 (IBM Crop., Armonk, NY) with the level of significance set at 0.05 (2-tailed).

As the unit of analysis for the associations between the practice environment and nurse-sensitive outcomes was at ward level, a total of 71 wards were included in the analysis, sufficient to detect a correlation coefficient as small as 0.33 with 80% at a 2-sided 5% level of significance.

3 | Results

3.1 | Sociodemographic Characteristics

The number of nurse participants in each wave of the surveys ranged from 604 to 865. Most (> 70%) of the participants

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were female and aged 45 or below (> 80%). The majority of participants (> 80%) were registered nurses (RNs), and over 10% were advanced practice nurses. The remaining participants consisted of ward managers, nursing officers, nurse specialists and nurse consultants. The responsibilities of nurse consultants in Hong Kong include delivering nurse-led care, establishing nursing standards and care protocols, and leading clinical and research initiatives to enhance patient care. Over 70% of participants spent 76%–100% of their working time on direct patient care duties. Most participants had either a Diploma or a Bachelor's degree at entry as an RN, with only a minority (< 4%) receiving a Master's degree prior to registration. A Bachelor's degree was the highest educational attainment for a majority of participants, followed by a Master's degree. Participants' experience in their current positions ranged from 3.7–4.1 years, while years of experience as an RN ranged from an average of 5–5.7 years. The average working duration in their current ward ranged from 28 to 36 months. Table 1 contains a summary of participants' sociodemographic information.

3.2 | Nurse Practice Environment

The composite and subscale scores of the PES-NWI are reported in Table 2. On the whole, the nurse practice environment was rated favourably, with a composite score of 53.5 out of 100. All subscales scored above 50 with the exception of the 'Staffing and resource adequacy' subscale which scored 36.7. The most favourable subscale was 'Nursing foundations for quality of care', which scored 59.6.

3.3 | Associations Between Nurse-Sensitive Patient Outcomes and Practice Environment

No significant associations were found between the overall nurse practice environment and the four nurse-sensitive patient outcomes. However, the subscale of 'staffing and resource adequacy' was significantly associated with both the outcomes of number of fall incidence and number of patients with CAUTIs. Higher levels of staffing and resource adequacy would be significantly associated with a lower number of fall incidence and a lower number of patients with CAUTIs. The results are reported in Table 3.

4 | Discussion

Previous studies suggest that nurses' work environments affect care quality and patient outcomes, with higher-rated practice environments associated with reduced odds of adverse events (Lake et al. 2019). This is the first study to investigate associations between nurses' practice environment and nurse-sensitive patient outcomes in Hong Kong. Our results indicate that nurses' perceptions of the practice environment are consistent with global evidence, and that while the overall work environment did not have significant associations with any of the four nurse-sensitive patient outcomes, the subscale of staffing and resource adequacy was significantly associated with inpatient falls and CAUTI incidence.

Nurses' evaluations of their work environment were positive on the whole, though the subscale of 'staffing and resource adequacy' received a notably poor rating, reflecting dissatisfaction with the prevailing issue of staff shortages. Being overworked as a result of insufficient staff is a frequent grievance experienced by nurses, with a previous study conducted in a similar sample working in public hospitals finding excessive workloads to be associated with turnover intention (Leung et al. 2020). Higher nurse-to-patient ratios may also increase job dissatisfaction and burnout (Shin et al. 2018). Furthermore, the subsequent rationing of patient care in favour of clinical tasks can be detrimental to patients' health and experiences, causing higher rates of pressure injuries, falls and nosocomial infections (Papastavrou et al. 2014). Appropriate staffing and retention policies should therefore be implemented to address the issue of nurse shortages so as to alleviate workforce concerns and facilitate higher standards of nursing care.

In terms of the effect of the practice environment on patient outcomes, better perceptions of staffing and resource adequacy were significantly associated with reduced incidence of both falls and CAUTIs. In particular, as there is a paucity of evidence regarding the association between inpatient falls and the nurse practice environment, this is a notable finding consistent with studies which suggest that better staffing as well as favourable nurse views of staff availability are linked with fewer inpatient falls (Al-Ghraiybah et al. 2021; Lee and Scott 2016) Aspects of nursing practice such as situational awareness, conducting risk assessment and providing education to patients and family carers—tasks often dependent on staff availability—have been identified as factors affecting inpatient fall incidence, highlighting the necessity of enhancing staffing to ensure that nurses feel empowered to make optimal care decisions when addressing the risk of inpatient falls (Takase 2022) Similarly, nurses play a central role in the mitigation of CAUTIs, with a systematic review of 10 studies involving over 25,000 patients indicating that nurse-driven protocols significantly reduced both catheter utilisation and CAUTI incidence (Su 2025) The presence of more nursing staff with higher qualifications and specialty certifications has also been associated with lower CAUTI incidence, reflecting our finding and suggesting that nurses within an adequately-staffed work environment may feel more confident in carrying out urinary catheter care according to best practices (Park et al. 2018).

On the other hand, no significant association between the overall nurse practice environment and nurse-sensitive patient outcomes was found, indicating that the relationship may be mediated by other nurse factors. For example, features of the study sample such as strong nurse resilience could be a potential mediating factor mitigating the impact of the practice environment on patient outcomes. Particularly amidst the COVID-19 pandemic but even in previous years, there has been evidence demonstrating the role of resilience in reducing adverse environmental impacts on both nurse and patient

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TABLE 1 | Characteristics of the nurse participants across four survey waves.

Characteristics	Survey wave			
	First ($n = 865$)	Second ($n = 604$)	Third ($n = 680$)	Fourth ($n = 661$)
Sex				
Female	74.3%	74.4%	78.2%	75.9%
Male	25.7%	25.6%	21.8%	24.1%
Age (years)				
≤25	21.1%	22.5%	19.9%	22.8%
26–30	31.5%	34.7%	35.8%	37.1%
31–35	11.9%	10.6%	10.5%	10.4%
36–40	10.1%	10.9%	12.7%	10.4%
41–45	6.9%	8.1%	9.1%	8.1%
46–50	5.6%	7.6%	6.9%	6.8%
51–55	3.9%	4.0%	4.2%	3.2%
56–60	1.1%	1.6%	0.9%	1.0%
>60	7.9%	0.0%	0.0%	0.3%
Current position				
RN	82.1%	83.3%	82.3%	82.5%
APN	13.0%	12.4%	13.7%	13.8%
NO	1.4%	1.8%	1.3%	1.0%
NS	0.4%	0.7%	0.6%	0.7%
WM	3.1%	1.8%	1.9%	1.8%
NC	0.0%	0.0%	0.2%	0.2%
Average working time on direct patient care				
51%–75%	30.0%	29.3%	29.3%	29.5%
76%–100%	70.0%	70.7%	70.7%	70.5%
Educational level at entry of RN				
Diploma	49.1%	50.5%	50.0%	47.3%
Bachelor	47.9%	46.5%	46.7%	50.9%
Master	3.0%	3.0%	3.3%	1.8%
Highest educational attainment				
Diploma	13.3%	14.6%	12.2%	13.4%
Bachelor	60.6%	60.4%	61.8%	60.5%
Master	26.1%	25.0%	25.0%	24.8%
Doctorate	0.0%	0.0%	0.8%	1.3%
Years of experience on current position†	4.0 (2.0–8.3)	4.1 (1.7–9.0)	3.9 (1.9–8.0)	3.7 (1.9–7.4)
Years of experience as registered nurse†	5.7 (2.4–14.0)	5.0 (2.0–14.3)	5.4 (2.1–15.0)	5.3 (2.3–14.7)
Working duration in current ward (months)†	36 (12–84)	28 (9–79)	33 (12–84)	30 (12–85)

Note: Data of variables marked with '†' are presented as median (inter-quartile range), all others are presented as %.
Abbreviations: APN, advanced practice nurse; NC, nurse consultant; NO, nursing officer; NS, nurse specialist; RN, registered nurse; WM, ward manager.

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TABLE 2 | Practice environment of the study wards.

	Mean (SD)	Inter-quartile range
Practice environment subscales (PES-NWI)[a] [Monthly average]		
Nurse participation in hospital affairs	53.3 (6.6)	48.0–57.9
Nursing foundations for quality of care	59.6 (4.8)	56.6–62.8
Nurse manager ability, leadership, & support of nurses	59.0 (8.6)	54.6–64.8
Staffing and resource adequacy	36.7 (7.8)	30.7–42.7
Collegial nurse-physician relations	59.0 (5.6)	55.6–62.5
Composite score	53.5 (5.7)	49.3–57.0

Abbreviation: PES-NWI, The Practice Environment Scale of the Nursing Work Index.
[a]The possible ranges of the five domain and composite scores of the PES-NWI were rescaled from the range of 1–4 to the range of 0–100.

outcomes (Gensimore et al. 2020). Moreover, a recent study on 270 frontline nurses from the Philippines revealed that resilience was a strong mediating factor which limited the harmful impacts of compassion fatigue on nurses' quality of care and job satisfaction (Labrague and de Los Santos 2021). Similarly, though our findings suggest that aspects of the practice environment such as staffing and resource adequacy were perceived poorly by nurses, that the overall work environment was not found to have a significant association with patient outcomes may imply that participating nurses had high resilience and were able to cope with environmental difficulties to moderate potential negative effects on patient health. Further investigations should therefore be carried out to explore the role of associated nursing practice factors such as resilience, burnout levels, or workload in mediating the impact of the nurse practice environment on patient outcomes in Hong Kong.

Moreover, this non-significant result may also be a result of several key differences in the design of the study, including the types of wards recruited, the prospective approach, and types of patient outcomes studied. For instance, while previous evidence has been based on data collected from a variety of units across hospitals including emergency, intensive care, mother and baby units and more (Cho et al. 2016; Hall et al. 2016), ours focused on adult medical, general surgical and orthopaedic wards only. As the working environment can vary significantly depending on the unique demands of each ward, this could be a reason for the contrasting findings. Several systematic and literature reviews also indicate that most studies investigating the association between the practice environment and patient outcomes are cross-sectional and cannot take variations in environmental or nurse factors into account (Stalpers et al. 2015; Al-Ghraiybah et al. 2021; Bae 2011).

Additional prospective studies with a longer period of investigation conducted in a variety of settings are also urgently required to strengthen the evidence on the impact of the nurse practice environment on patient outcomes. Findings from the present study reinforce the importance of optimising nurse staffing as a pressing concern and provide directions for future investigations, such as using structural models to explore mediating pathways to better understand specific mechanisms behind the effects of the nurse working environment on patient outcomes. Staffing policies which have been effective in other regions around the world, such as the introduction of safe nurse–patient ratios (Driscoll et al. 2018), should also be tested and implemented to improve working conditions and empower nurses to maintain a high standard of patient care.

4.1 | Limitations

This study has some limitations. As only nurses working in adult medical, general surgical and orthopaedic wards from public hospitals were recruited, findings may not be generalisable to other settings where nurses' working conditions may differ. Moreover, the methods of data collection may have also resulted in underreporting of adverse events, particularly regarding physical restraint use and CAUTIs. As data on the daily usage of physical restraint devices and catheters was not available in a manner accessible to the research team, collaboration with infection control nurses was required to screen a vast database of specimen results, patient medical records and medical diagnoses in order to obtain the required data. Any potential gaps in data collection and consolidation may have therefore affected the study results on physical restraint use and CAUTIs.

5 | Conclusion

Our findings suggest that while the overall nurse practice environment was not significantly associated with nurse-sensitive patient outcomes, the subscale of staffing and resource adequacy was significantly associated with the outcomes of inpatient falls and CAUTIs. We suggest that nurse resilience may be a mediating factor limiting the impact of the practice environment on patient outcomes, and further studies should be conducted to investigate this potential relationship. Additional directions for research also include testing effective strategies to promote practice environments that deploy adequate nurses and resources for enhanced quality of nursing care.

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TABLE 3 | Associations between nurse-sensitive patient outcomes and practice environment.

Practice environment subscales (PES-NWI) [a]	Nurse-sensitive patient outcomes							
	Number of patients with pressure injuries†		Number of fall incidence†		Number of physical restraints		Number of patients with CAUTI†	
	B (95% CI)	*p*	*B* (95% CI)	*p*	*B* (95% CI)	*p*	*B* (95% CI)	*p*
Nurse participation in hospital affairs	−0.002 (−0.020, 0.017)	0.839	0.001 (−0.015, 0.017)	0.894	1.09 (−1.74, 3.92)	0.443	0.006 (−0.009, 0.022)	0.412
Nursing foundations for quality of care	−0.008 (−0.033, 0.017)	0.516	−0.004 (−0.025, 0.017)	0.715	0.62 (−3.10, 4.34)	0.740	0.005 (−0.016, 0.025)	0.646
Nurse manager ability, leadership, & support of nurses	0.000 (−0.014, 0.013)	0.942	0.001 (−0.011, 0.012)	0.881	1.01 (−1.00, 3.03)	0.319	0.009 (−0.002, 0.020)	0.114
Staffing and resource adequacy	−0.009 (−0.023, 0.006)	0.229	−0.013 (−0.024, −0.001)	0.037	−1.00 (−3.19, 1.18)	0.363	−0.012 (−0.024, −0.001)	0.032
Collegial nurse-physician relations	0.002 (−0.018, 0.023)	0.809	0.001 (−0.017, 0.018)	0.947	0.37 (−2.77, 3.50)	0.816	−0.001 (−0.018, 0.017)	0.945
Composite score	−0.005 (−0.026, 0.016)	0.625	−0.005 (−0.023, 0.013)	0.558	0.54 (−2.68, 3.75)	0.741	0.001 (−0.017, 0.019)	0.909

Note: The outcomes marked with '†' were square root transformed before being subjected to association analysis. B: regression coefficient estimated by multiple regression with adjustment for hospital; it represents the average number of each (transformed) outcome increased when the rescaled subscale or composite score of PES-NWI is increased by 1%.
Abbreviation: PES-NWI, The Practice Environment Scale of the Nursing Work Index.
[a]The subscale and composite scores of PES-NWI were rescaled to the range of 0–100 before being subjected to association analysis.

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Author Contributions

J.P.C.C. and S.H.S.L. contributed to the study conceptualisation, methodology, funding acquisition, investigation, analysis, participant recruitment, data collection and drafting of the manuscript. K.C.C. contributed to the methodology, analysis and drafting of the manuscript. D.W.K.T. and A.M.L.K. contributed to participant recruitment and data collection. L.B. and D.R.T. contributed to the analysis and drafting of the manuscript. E.L.S.C., I.F.K.L., D.T.F.L. and M.D.M. contributed to the design of the study. All authors read and approved the final manuscript.

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Disclosure

Statistics: There is a statistician on the author team (Kai Chow CHOI).

Ethics Statement

Ethical approval was obtained from the Joint CUHK-New Territories East Cluster Clinical Research Ethics Committee and the Kowloon West Cluster Research Ethics Committee (CRE-2013.101).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated during and/or analysed during the current study are not publicly available, but are available from the corresponding author on reasonable request.

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